

Non-Linear Dynamics of Government Debt and Economic Growth: Evidence from Canada, 1990-2023

Chris Perfitt

ECON4010-Y

Dr. Shutao Cao

Trent University

Date: TBD

Abstract:

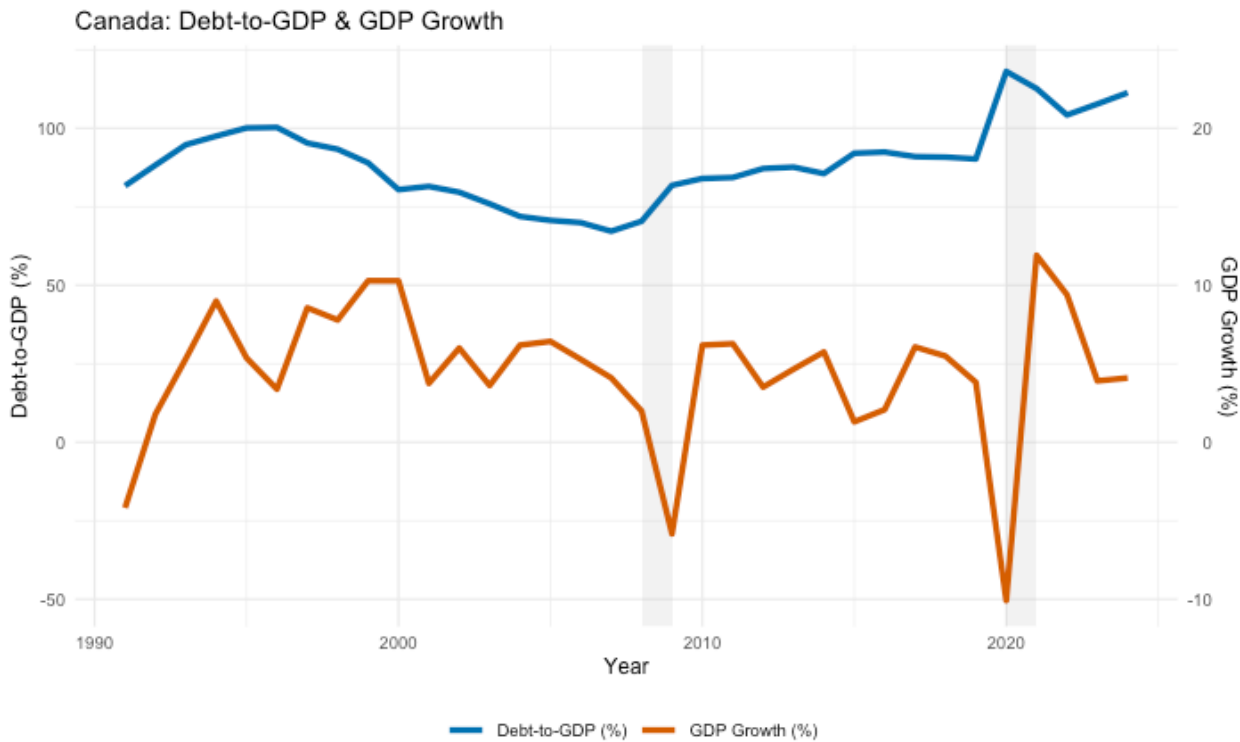
The relationship between government debt and economic growth remains highly debated in both academic research and political circles. Reinhart and Rogoff (2010) have argued that this relationship follows an inverted U-shape relationship, implying that moderate debt supports growth, but excessive debt becomes debt-restrictive. This paper tests whether Canada follows the same pattern using annual data from 1990 to 2023 (STATSCAN). This was done by estimating a quadratic OLS regression with lagged debt specifications, productivity controls, and crisis-period dummy variables. This was done to capture the potential non-linear and state-dependent effects. Once these regressions were completed, the results showed a contradictory story compared to theoretical predictions. A U-shaped association in the baseline specification was found, where the debt-growth relationship is minimized at 75-79% debt-to-GDP. However, this relationship differed across different macroeconomic regimes. During the 2008-09 financial crisis, a higher lagged debt is associated with stronger growth (marginal effect = 0.11, $p < 0.05$), which is consistent with counter-cyclical fiscal policy. But during the 2020-21 COVID pandemic, this relationship reverses (marginal effect = -0.02, $p < 0.01$). These findings suggest that the debt-growth relationship may be state-dependent rather than a structural indicator for GDP growth.

I. INTRODUCTION:

Does government debt constrain economic growth? This question is what pushed me to examine the data and draw conclusions. The issue with this is that this is fundamentally challenged since the debt-to-GDP ratio is affected by the GDP growth through the denominator, making it difficult to isolate the effect that debt has on growth. When economic growth slows, the previously mentioned ratio will automatically rise even without additional borrowing. This paper addresses this challenge by examining whether lagged government debt predicts future economic growth. This was used due to the behaviour of debt; it is slow-moving, and it may take more time to see the impact. Therefore, the use of lagged debt to examine the previous year's debt-to-GDP ratio on the current year's GDP growth. Using annual data from 1990 to 2023, I estimate the relationship between the past year's debt level and current GDP growth while controlling for productivity shocks, and testing for heterogeneity across different macroeconomic regimes (2008-09, 2020-21). By focusing on lagged specifications, I mitigate the concerns about the automatically rising ratios while examining whether fiscal conditions in one period may influence growth outcomes in the next.

This research question is particularly interesting in the Canadian context because this economy has been through two major economic crises in the last three decades. See Table A1 in the appendix for full data which shows how Canada's general government debt-to-GDP ratios have changed through the years that this paper looks at. In the 1990's, Canada's debt ratio was between 74% to 80%, although in 1995, this ratio was 100%. Then in the 2000's, it bounced around 80% to 60%. After the 2008 crisis, the debt has been steadily increasing until it hit its peak within the last 33 years at 118%. Due to this fact, the policymakers will be forced to make decisions while juggling competing priorities such as maintaining investment, healthcare, infrastructure, and education, all while managing debt sustainability (Department of Canada, 2024). Canada's experience with these two different crises (demand-side shock of 2008-09 and supply-side shock of 2020-21) allows for an interesting setting to investigate the impact that a lagged debt-to-GDP ratio has on Canada's future economic growth.

Figure 1: Canada's Debt-to-GDP Ratio and GDP Growth, 1990-2023



Note: Shaded regions indicate crisis periods (2008-09 global financial crisis, 2020-21 COVID-19 pandemic). GDP Growth is scaled by a factor of 10 for dual-axis visualization. *Source:* Debt data from IMF World Economic Outlook Database (2024); GDP data from Statistics Canada.

I find evidence of a non-linear, state-dependent relationship. In the baseline specifications, the association between lagged debt and growth follows a U-shape, minimized at approximately 75-79% debt-to-GDP. However, this average relationship hides a massive variation across crisis periods. During the 2008-09 financial crisis, higher lagged debt is strongly associated with subsequent growth (marginal effect = 0.11, $p < 0.05$), consistent with counter-cyclical fiscal expansion that supports recovery after this crisis. However, during the 2020-21 COVID pandemic, this relationship reverses (marginal effect = -0.02, $p < 0.01$), suggesting fiscal constraints or different crisis dynamics. These findings contribute to debt-growth literature in two distinct ways. First, using lagged specifications, I give evidence that reduces some of the reverse causality bias compared to cross-sectional analyses. Second, I demonstrate that the debt-growth relationship is state-dependent rather than structural, meaning that the same level of debt may support or constrain economic growth depending on the type of macroeconomic shocks. This has clear implications for fiscal policy construction, suggesting that debt sustainability assessments should focus on specific economic conditions rather than a certain universal threshold.

The remainder of this paper is organized as follows. Section 2 will review the empirical literature on debt and growth with strong attention to the threshold framework. Section 3 presents the theoretical framework, which shows the linkage of debt to economic performance. Section 4 will describe the data sources and the empirical methodology, including the lagged debt and the crisis variables. Section 5 will present the results and give an

analysis of those results. Section 6 will discuss the limitations and the implications. Finally, section 7 will conclude.

II. LITERATURE REVIEW:

Within the economics discipline, the empirical relationship between government debt and economic growth has been researched extensively, but the consensus has been mixed. The results of different papers vary widely across different studies. The main debates in this space look at whether a debt threshold exists, the direction of causality, and whether debt affects growth systematically or the relationship reflects reverse causality. Methodologies of the different literature either look at a single country time series, or some papers use cross-country panels, which leads to differing results. These issues have allowed the discussion to evolve and differ across different economists. This literature review will summarize the key findings, point out the methodological challenges, and identify the gaps which influenced the present analysis.

The first and most influential piece of literature that will be analyzed is Reinhart and Rogoff's 2010 paper: *Growth in a Time of Debt*. The center of their paper is that the median growth drops when the debt-to-GDP ratio falls after hitting a debt threshold of 90%. This data was found by using 44 countries over 200 years of data. To further look at their threshold claim, they claimed that after a country's debt-GDP ratio had 90%, they saw a median decrease in GDP growth. To further this point, they explain how a country will not simply grow their way out of these high-debt levels. After publishing this paper, the impact was immediate and powerful, as policymakers were citing this paper as a reference to allow them to argue for steep reductions in budget deficits. This was known as Austerity Justification, and European policymakers used their paper to push for this reduction in budget deficits. Not only was this used to justify Austerity, but it was also used by Olli Rehn, the European Commissioner for the Economy, to argue that high public debt was acting as a "permanent drag on growth" (Rehn, 03/07.2013). This spearheaded a drastic policy change that was aimed at reducing fiscal budget deficits. As shown, this paper vastly influenced European policy during the economic crisis. However, there was an issue. Herndon et al. 2014 found coding errors within their calculations, which led to a large and ongoing debate on whether a debt threshold exists. After the coding error has been examined, the 90% threshold largely disappeared. Now, the threshold has been questioned, but most agree that the relationship between debt and growth follows non-linear behaviour.

Panizza and Presbitero published "Public Debt and Economic Growth in Advanced Economies" in 2013, which focuses on the issue of reverse causality that comes from using the debt-to-GDP ratio. This issue arises when the GDP growth slows; the debt ratio will rise even though there was no additional borrowing, which creates a simultaneity bias. They argue that this is extremely difficult to isolate causal direction, and cross-country analysis is even more susceptible to being influenced by this issue. The difficulty arising from this problem can

be mitigated by using a lagged debt-growth relationship, but this does not solve everything. Without addressing this issue, it becomes impossible to determine the impact that high government debt has on economic growth. Therefore, this paper used the lagged debt-growth ratio to attempt to mitigate the reverse causality effect that this specific ratio is prone to.

While the debate continues about specific thresholds, there is a strong consensus that the debt-growth relationship is nonlinear. Checherita-Westphal & Rother (2012) have found evidence of the inverted U-shape, just as Reinhart & Rogoff found, where the marginal effects of debt depend very heavily on existing debt levels. However, in this study, the estimated turning points vary vastly over the study; these range from 60% to over 100% of GDP. This implies that the relationship is heterogeneous rather than being a universal threshold, which is what Reinhart & Rogoff's paper argued. This research found variation between different European countries as they have different characteristics, such as different monetary policy frameworks. Yet there has been very little focus paid to whether the debt-growth relationship varies by the different types of economic shocks.

Therefore, even after the extensive research on debt thresholds and whether this relationship follows a linear or nonlinear behaviour, many gaps remain. First, much of the existing evidence relies on cross-country panel data, which is faced with reverse causality. Second, while the nonlinear relationships have been documented, relatively little attention has been paid to whether the debt-growth relationship has varied systematically across different macroeconomic regimes within a single G7 country. Therefore, this paper addresses these gaps by estimating a nonlinear, lagged time-series model for Canada, explicitly incorporating production controls and crisis-period interactions. Since this paper focuses solely on Canada and controlling the variation across different macroeconomic regimes, the analysis can give insight into whether the debt-growth relationship reflects a structural threshold or is a state-dependent association that varies with economic conditions.

III. DATA

This paper uses annual Canadian time-series data covering the period 1990-2023. This sample window was chosen by joint data availability across all variables included in the empirical specifications:

The primary variables are:

→ Real GDP Growth (%), measured as the annual percentage change in real GDP with 2017 as the reference year (Statistics Canada).

→ General Government Gross Debt, obtained from the IMF World Economic Outlook database.

→ Government Debt-to-GDP ratio (%), calculated using the two previous primary variables.

→ Multifactor Productivity (MFP), sourced from Statistics Canada.

→ Crisis Period Dummy Variables for:

- 2008-09 Global Financial Crisis
- 2020-21 COVID-19 Pandemic

Debt is expressed in levels of as a percentage of Canada's GDP. Output and productivity variables enter the model in growth rates. Productivity is primarily specified as the first difference of log MFP, $\Delta \log(MFP)$, to capture short-run technological shocks while mitigating concerns related to productivity levels.

The effective sample size ranges from 32 to 33 observations, depending on the lag structure and model specification.

IV. 1- BASELINE EMPIRICAL SPECIFICATION:

To evaluate whether the relationship between government debt and economic growth is nonlinear, I estimate the following quadratic time-series regression:

$$g_t = \alpha + \beta_1 Debt_t + \beta_2 Debt_t^2 + \gamma \Delta \log(MFP)_t + u_t$$

Where:

- g_t denotes real GDP growth in year t,
- $Debt_t$ is the government debt-to-GDP ratio,
- $\Delta \log(MFP)_t$ captures the productivity growth,
- u_t is the error term

The quadratic specification allows for potential non-linear effects of government debt to be identified. The standard theory argues that there is an inverted U relationship, meaning that low levels of debt may be growth-enhancing while excess debt may be a limitation. However, the data may also reveal alternative patterns. The implied turning points were found with:

$$Turning\ Point = -\frac{\beta_1}{2\beta_2}$$

IV.2 - BASELINE EMPIRICAL RESULTS:

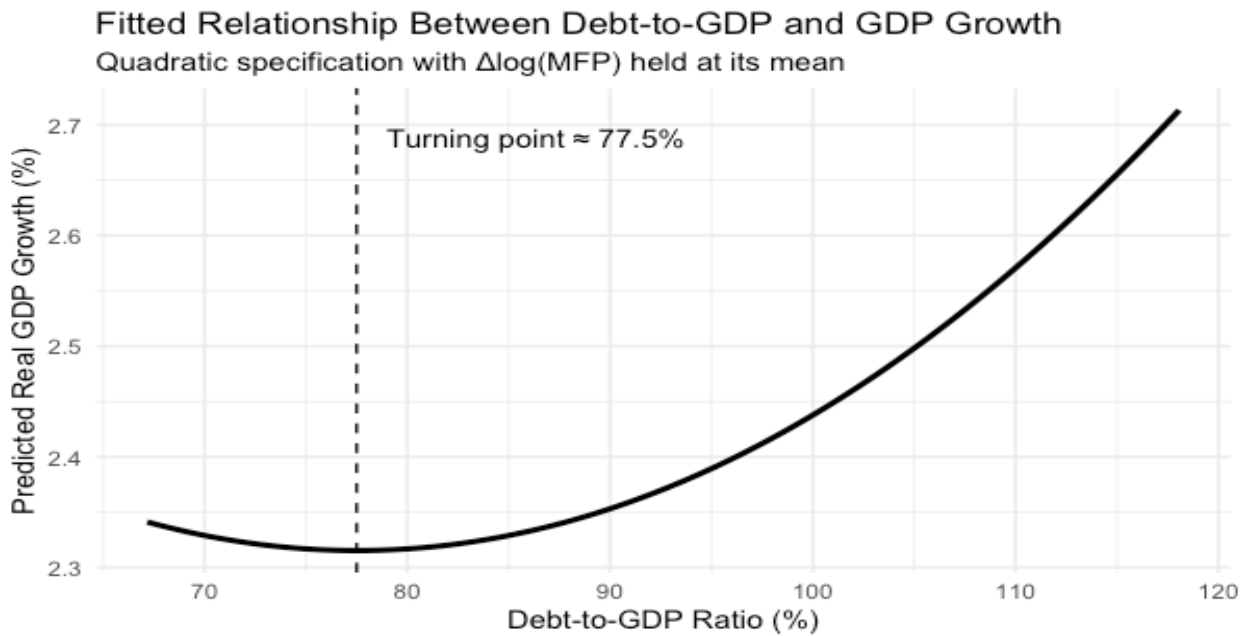
Table B1: Canada — Baseline GDP Growth Regressions (No Lag)

Variable	Linear (1)	Quadratic (2)
Debt-to-GDP (t)	0.006*** (0.002)	-0.037* (0.019)
Debt-to-GDP ² (t)	—	0.0002** (0.0001)
$\Delta \log(MFP)$	1.013*** (0.010)	1.020*** (0.010)
Constant	-0.512*** (0.154)	1.426* (0.832)

Observations	33	33
R ²	0.997	0.997
Adjusted R ²	0.997	0.997
Turning Point (Debt-to-GDP %)	—	77.51%

Note: Dependent variable is Real GDP Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. OLS estimation with Newey-West HAC standard errors.

Figure B1: Fitted Relationship Between Debt-to-GDP and GDP Growth (Baseline)



Note: Quadratic specification with $\Delta \log(MFP)$ held at its sample mean.

Above, the baseline regression implies that the relationship between the debt-to-GDP ratio follows a nonlinear behaviour, which is consistent with a relationship following a U-shape, as shown above in the graph. This corresponds with the computed turning point $\sim 77.51\%$, which represents the point at which the debt-to-GDP ratio is associated with GDP growth. This idea implies that at lower levels of debt, an increase in the debt-growth level accompanies with downturns. This allows for the idea that after the debt-to-GDP ratio reaches the turning point, the growth of GDP will increase. This pattern likely reflects the small data sample, which cluster around post-crisis recovery periods when the growth rebounded. But to test this idea, the need for crisis variables to be included is section VI. To further examine the impact that the debt-to-GDP ratio has on GDP growth, the idea of lagging variables must be introduced; this is due to the behaviour of government debt, as it is a non-volatile variable, which leads it to have a slower effect.

V.1 - LAG STRUCTURE AND REVERSE CAUSALITY MITIGATION:

The challenge of reverse causality arises from the relationship that is embedded in the debt-to-GDP ratio, this is since GDP shows up in the denominator, meaning that negative growth shocks that are exogenous will mechanically increase the ratio even though borrowing has not increased, causing there to be a bias error.

To deal with this as best as possible, the use of a lagged specification was needed:

$$g_t = \alpha + \beta_1 Debt_{t-1} + \beta_2 Debt_{t-1}^2 + \gamma \Delta \log(MFP)_t + u_t$$

and:

$$g_t = \alpha + \beta_1 Debt_{t-2} + \beta_2 Debt_{t-2}^2 + \gamma \Delta \log(MFP)_t + u_t$$

Where:

- g_t denotes real GDP growth in year t,
- $Debt_{t-1}$ and $Debt_{t-2}$ is the government debt-to-GDP ratio lagged by 1 and 2 years,
- $\Delta \log(MFP)_t$ captures the productivity growth,
- u_t is the error term

Lagging the debt variable helps isolate whether the prior fiscal conditions in the previous years are associated with subsequent GDP growth. The use of the lagged variable allowed the regression model to test whether debt levels exert a delayed influence on the current GDP growth rate, which is more consistent with the behaviour of the debt-to-GDP ratio. While this does not fully solve this problem, it weakens that effect.

V.2 – 1 YEAR LAG STRUCTURE RESULTS:

Table C1: GDP Growth with 1-Year Lagged Debt-to-GDP

Variable	Coefficient
Lagged Debt-to-GDP (t-1)	-0.038** (0.017)
Lagged Debt-to-GDP ² (t-1)	0.0003** (0.0001)
$\Delta \log(MFP)$	0.991*** (0.009)
Constant	1.433* (0.753)
Observations	33
R ²	0.998
Adjusted R ²	0.998
Turning Point (Debt-to-GDP %)	74.79%

Note: Dependent variable is Real GDP Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

The table examines the next step in the research, which is looking at the impact that the government debt-to-GDP ratio being lagged variable, which strengthens the interpretation of the relationship by reducing concerns within the OLS regressions. The continuity of the U-shape under lagged debt suggests that the observed relationship is not only driven by growth conditions. The result suggests that at lower levels of debt, an increase will be accompanied by a reduction in GDP growth. While considering the persistent U-shape relationship, the computed turning point of $\sim 74.89\%$ suggests that once the debt-to-GDP ratio reaches this point, it may indicate higher GDP growth. This implies that as debt is being accumulated, the macroeconomic policies may reflect expansionary policy responses. To further examine this relation, the next logical step is to look at a 2-year lagged model.

V.3 - 2 YEAR LAG STRUCTURE RESULTS:

Table C2: GDP Growth with 2-Year Lagged Debt-to-GDP

Variable	Coefficient
Debt-to-GDP (t-2)	-0.061*** (0.012)
Debt-to-GDP ² (t-2)	0.0004*** (0.0001)
$\Delta \log(\text{MFP})$	1.000*** (0.006)
Constant	2.418*** (0.531)
Observations	32
R ²	0.999
Adjusted R ²	0.999
Turning Point (Debt-to-GDP %)	79.12%

*Note: Dependent variable is Real GDP Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.*

Above, I estimate a new table that further extends the baseline and the lagged analysis by examining whether the debt-to-GDP ratio lagged by 2 years affects the current GDP growth. The persistent U-shaped relationship under the 2-year lagged specification suggests that this observed pattern is not driven solely by contemporaneous economic conditions. At the lower levels of debt, the increase in government debt is associated with weaker growth. The computed turning point of $\sim 79.12\%$ represents the level of 2-year lagged debt, at which the association with GDP is minimized, beyond which higher debt levels are associated with stronger growth outcomes. This continued pattern may reflect an accommodative macroeconomic policy and sustained fiscal capacity. To further assess the robustness of the relationship, the next step is to consider an interactive model.

VI.1 - STATE DEPENDENT INTERACTION MODEL:

To assess whether the debt-growth relationship changes across different macroeconomic regimes, I estimate an interactive model with lagged debt:

$$g_t = \alpha + \beta_1 Debt_{t-1} + \beta_2 Debt_{t-1}^2 + \gamma \Delta \log(MFP)_t + \delta D_{2008-09,t} + \delta D_{2020-21,t} \\ + \theta_1 (Debt_{t-1} \times \Delta \log(MFP)_t) + \theta_2 (Debt_{t-1} \times D_{2008-09,t}) + \theta_3 (Debt_{t-1} \times D_{2008-09,t}) + u_t$$

Where:

- g_t denotes real GDP growth in year t,
- $Debt_{t-1}$ is the government debt-to-GDP ratio lagged by 1 year,
- $\Delta \log(MFP)_t$ captures the productivity growth,
- $D_{2008-09,t}$ is a dummy variable for 2008-09, equals 1 for 2008-09 and 0 otherwise,
- $D_{2020-21,t}$ is a dummy variable for 2020-21, equals 1 for 2020-21 and 0 otherwise,
- u_t is the error term

This specification allows for the debt-growth relationship to vary along three different dimensions. The first being the quadratic term, this allows the nonlinear effects of debt that are dependent on the level of debt. Second, the interaction $Debt_{t-1}$ and $\Delta \log(MFP)_t$ allows the effect of debt to be different depending on productivity shocks. Lastly, the crisis interactions allow the debt-growth association to shift during the Global Financial Crisis and the COVID-19 Pandemic, respectively.

Since the debt enters both quadratically and through interaction terms, the marginal effects of debt are estimated:

$$\frac{\partial g_t}{\partial Debt_{t-1}} = \beta_1 + 2\beta_2 Debt_{t-1} + \theta_1 \Delta \log(MFP)_t + \theta_2 D_{2008-09,t} + \theta_3 D_{2020-21,t}$$

This expression implies that the marginal debt that is estimated is state-dependent. This means that it varies with the level of lagged debt, the productivity growth rate, and whether the economy is in a crisis. In particular, the model does not generate a single, fixed turning point, but rather, any found turning points are conditional on $\Delta \log(MFP)_t$ the crisis variables.

VI.2 - STATE DEPENDENT INTERACTION MODEL RESULTS:

Table D1: Interactive Model (Lagged Debt, ΔMFP , Crisis Dummies)

Variable	Coefficient
Debt-to-GDP (t-1)	-0.101*** (0.016)

Debt-to-GDP ² (t-1)	0.001*** (0.0001)
$\Delta \log(\text{MFP})$	1.215*** (0.132)
D(2008–09)	-6.665* (3.434)
D(2020–21)	2.487*** (0.698)
Debt(t-1) $\times$ $\Delta \log(\text{MFP})$	-0.002 (0.001)
Debt(t-1) $\times$ D(2008–09)	0.099* (0.051)
Debt(t-1) $\times$ D(2020–21)	-0.025*** (0.007)
Constant	3.817*** (0.761)
Observations	32
R ²	0.999
Adjusted R ²	0.999

Note: Dependent variable is GDP Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. HAC standard errors used throughout.

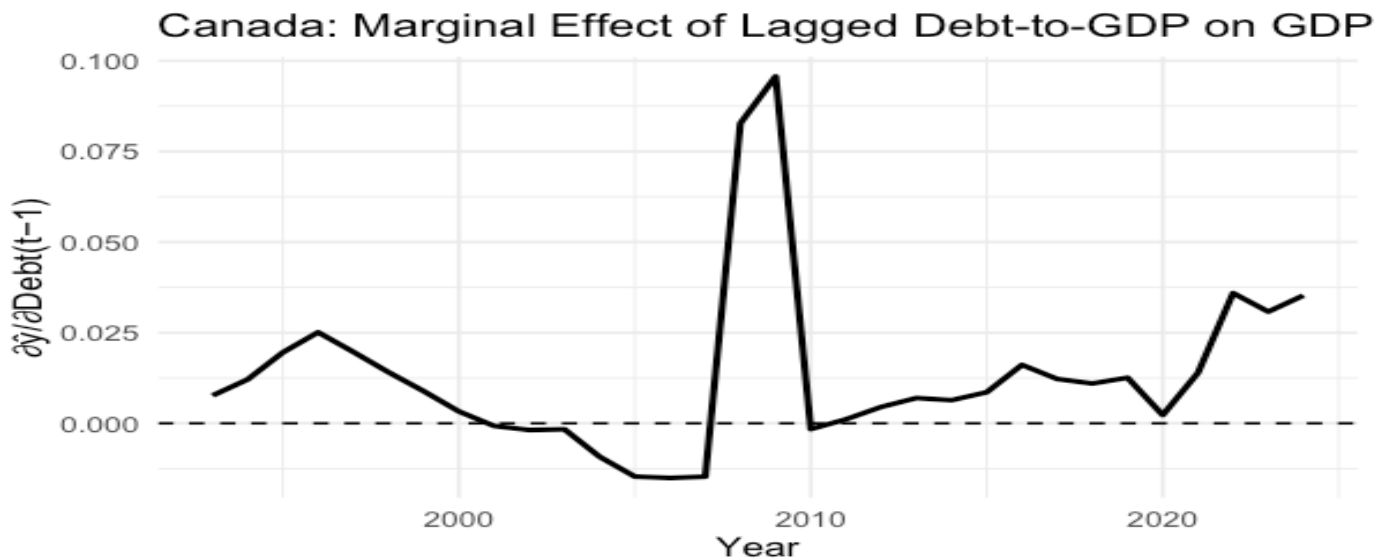
The table above examines an extension of the lagged version of the baseline model by adding interaction terms and dummy variables for two different crisis years. Allowing lagged debt to interact with the dummy variables of the Global Financial Crisis and the COVID-19 Pandemic suggests that the debt-growth relationship differs across different macroeconomic regimes. The interaction term with the 2008-09 variable is positive, while the interaction with the 2020-21 crisis is negative, implying that the association between debt accumulation and growth may depend on the nature of the shock.

Table D2: Marginal Effects of Lagged Debt-to-GDP on GDP Growth

Scenario	Marginal Effect	95% CI Lower	95% CI Upper
Normal year (means, no crisis)	0.0088	0.0036	0.0139
2008–09 crisis year (means)	0.1075	0.0560	0.1589
2020–21 crisis year (means)	-0.0166	-0.0258	-0.0074
Normal year ($\Delta \text{MFP} = 0$; no crisis)	0.0140	0.0047	0.0232

Note: Marginal effects computed using the delta method with HAC standard errors. Evaluated at sample mean of lagged debt-to-GDP.

Figure D1: Marginal Effect of Lagged Debt-to-GDP on GDP Growth (Year-by-Year)



To further interpret the nonlinear and interaction effects in the lagged debt model, the marginal effects of the lagged debt-to-GDP on GDP growth are evaluated under different macroeconomic environments. These marginal effects are computed at the sample mean level of the lagged debt and productivity growth; this allows the estimated association of debt and growth to be interpreted across different periods and episodes in the economic lifetime.

When looking at the scenario of normality in macroeconomic conditions, the marginal effect of lagged debt on growth is positive but modest; this is correlated with the interpretation of being at higher debt levels, additional borrowing is associated with a moderate improvement in growth outcomes. To examine the first Crisis year of 2008-09, the marginal effect of the lagged debt on growth is incredibly high; this is consistent with the theory that fiscal expansion plays a large role in stabilizing output when there is a large negative economic shock. Above, the graph of the marginal effects shows a large fluctuation during those specific years, which further implies the point that accumulating additional debt may have played a role in stabilizing the crisis. When looking at the marginal effects of the lagged debt-to-GDP growth ratio during the COVID-19 Pandemic, the contrast is stark. The marginal effect is negative, which suggests that the debt-growth relationship can be different across crisis types, which can lead to the potential reflection of supply-side shocks and/or non-economic constraints. When examining the marginal effects when the change of MFP is set to zero, the model yields a similar positive association, implying that the estimated debt effects are not driven solely by contemporaneous productivity shocks. When taken all together, the results suggest that there is a marginal association between government debt and GDP growth and that it may be state-dependent, varying across the different macroeconomic environments.

VII. STATIONARITY AND INFERENCE:

Since my research project relies on annual time-series data, I test for unit roots using the Augmented Dickey-Fuller (ADF) test. The results are in the appendix, and they indicate that debt-to-GDP levels exhibit

persistence that is consistent with non-stationarity and that GDP growth and productivity growth are stationary. Given that GDP Growth (the dependent variable) is stationary and productivity, the inclusion of quadratic terms and productivity controls further reduces the likelihood of spurious correlation. Nevertheless, my inference is done with caution in mind.

As well, all my regressions are estimated using OLS with Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors to allow for the general forms of these to be seen in the process. Diagnostic tests include:

- Breusch-Pagan test for heteroskedasticity,
- Breusch-Godfrey test for serial correlation
- Ramsey RESET test for function form misspecification

These diagnostics are reported in the appendix.

VIII.1 - FURTHER CONSIDERATIONS – GDP/CAPITA:

To tell a more coherent story, an alternative route was taken, which was looking at how the debt-to-GDP ratio would affect the GDPPC. GDP is very important to a country, but GDP per capita looks at a more personal level of how the government debt may affect the broader public. Therefore, this regression was conducted:

$$GDPPC = \alpha + \beta_1 Debt_t + \beta_2 Debt_t^2 + \Delta \log MFP + \delta D_{2008-09,t} + \delta D_{2020-21,t} + u_t$$

- $GDPPC$ denotes real GDP/capita growth in year t ,
- $Debt_t$ is the government debt-to-GDP ratio,
- $\Delta \log(MFP)_t$ captures the productivity growth,
- $D_{2008-09,t}$ is a dummy variable for 2008-09, equals 1 for 2008-09 and 0 otherwise,
- $D_{2020-21,t}$ is a dummy variable for 2020-21, equals 1 for 2020-21 and 0 otherwise,
- u_t is the error term

VIII.2 - FURTHER CONSIDERATIONS – GDP/CAPITA RESULTS:

Table E1: GDP per Capita Growth — Baseline Model

Variable	Base Model
Debt-to-GDP	0.202*** (0.052)
Debt-to-GDP ²	-0.001*** (0.0003)
$\Delta \log(MFP)$	1.032*** (0.025)

Dummy: 2008–2009	-0.003 (0.209)
Dummy: 2020–2021	1.893*** (0.279)
Constant	-9.049*** (2.246)
Observations	33
R ²	0.989
Adjusted R ²	0.987
Turning Point (Debt-to-GDP %)	80.04%

Note: Dependent variable is GDP per Capita Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

To tell a more grounded story, the next important step in the research was to look at the GDP / capita growth rate and examine how the debt-growth ratio affects this. The results show that this nonlinear relationship behaves in an exact opposite way to GDP growth, as it signifies an inverted U-shape. This implies that as the debt-to-GDP ratio increases, the GDP / capita growth will be associated with a higher outcome until said ratio reaches the turning point. The computed turning point is ~80.04%, meaning that before that point, as the debt-to-GDP ratio increases, GDP / capita will follow a procyclical relationship, implying that this result is associated with higher GDP / capita growth. Once this relationship reaches the ~80.04%, it may indicate lower GDP / capita growth.

IX. CONCLUSION:

This research paper set out with the goal of determining the effect that the debt-to-GDP ratio has on GDP growth while using Canada as its primary focus. The research used a nonlinear regression framework that incorporated quadratic debt terms, lagged debt specifications, productivity controls, and crisis-period dummy variables. After this was completed, the empirical results suggest that the debt-to-GDP relationship in Canada is neither strictly linear nor characterized by a universal threshold (which was suggested by Reinhart and Rogoff). Instead, the findings indicate a nonlinear and state-dependent association between government debt and economic performance.

Across the estimated regressions, the estimated relationship follows a U-shape, with the turning points between approximately 75% and 79% of GDP. These results suggest that an increase in government debt at relatively low levels is associated with weaker subsequent growth, which, once the ratio reaches the turning point, GDP growth becomes greater. However, when the crisis dummy variables are introduced, the relationship that was determined was found to mask significant variation across the different macroeconomic results. During the 2008-09 Global Financial Crisis, higher lagged debt was associated with stronger subsequent growth, which is consistent with counter-cyclical fiscal policy. But in contrast, in the Covid-19 pandemic, the marginal effect was

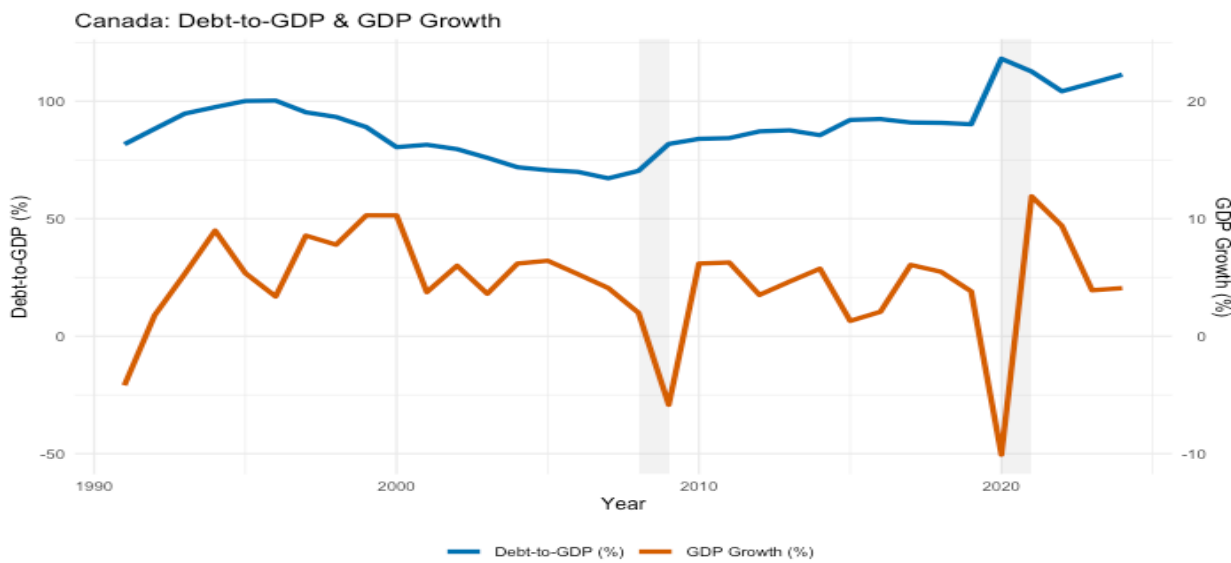
negative, which implies that the debt-to-GDP ratio affects the GDP growth differently depending on the nature of the economic shock.

After all has been said, these findings suggest that the debt-to-GDP ratio should not be evaluated solely through the fixed debt thresholds. Instead, the macroeconomic environment and the nature of the economic shocks that countries face play an important role in determining how fiscal policy interacts with economic performance.

X. APPENDIX:

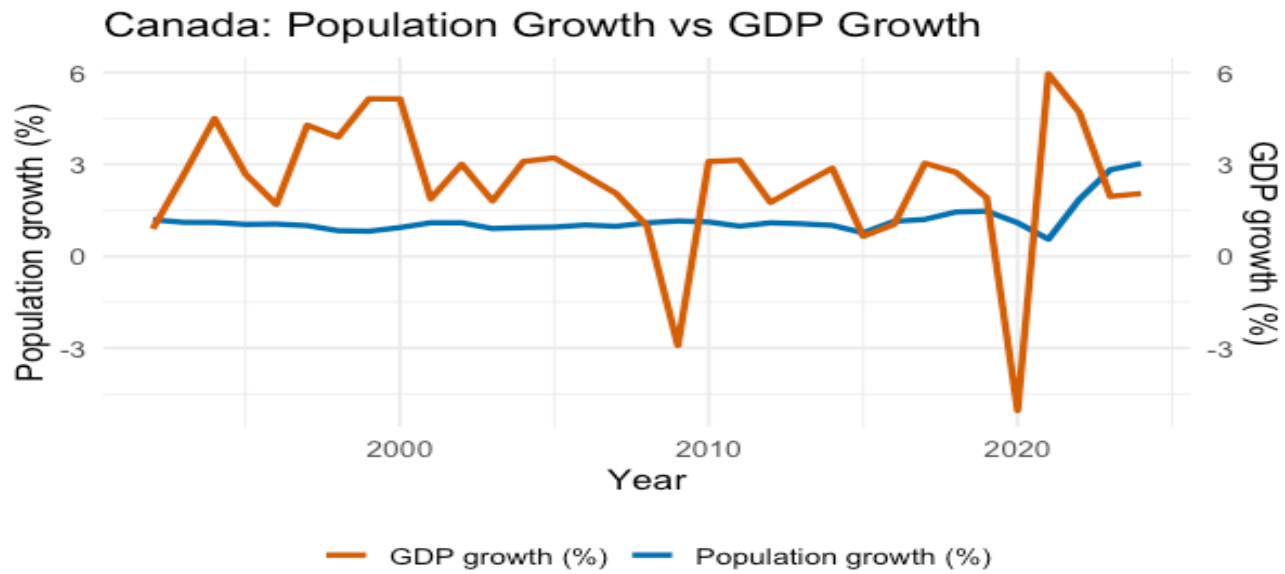
A. Descriptive Time Series Figures:

Figure A1: Canada's Debt-to-GDP Ratio and GDP Growth, 1990–2023



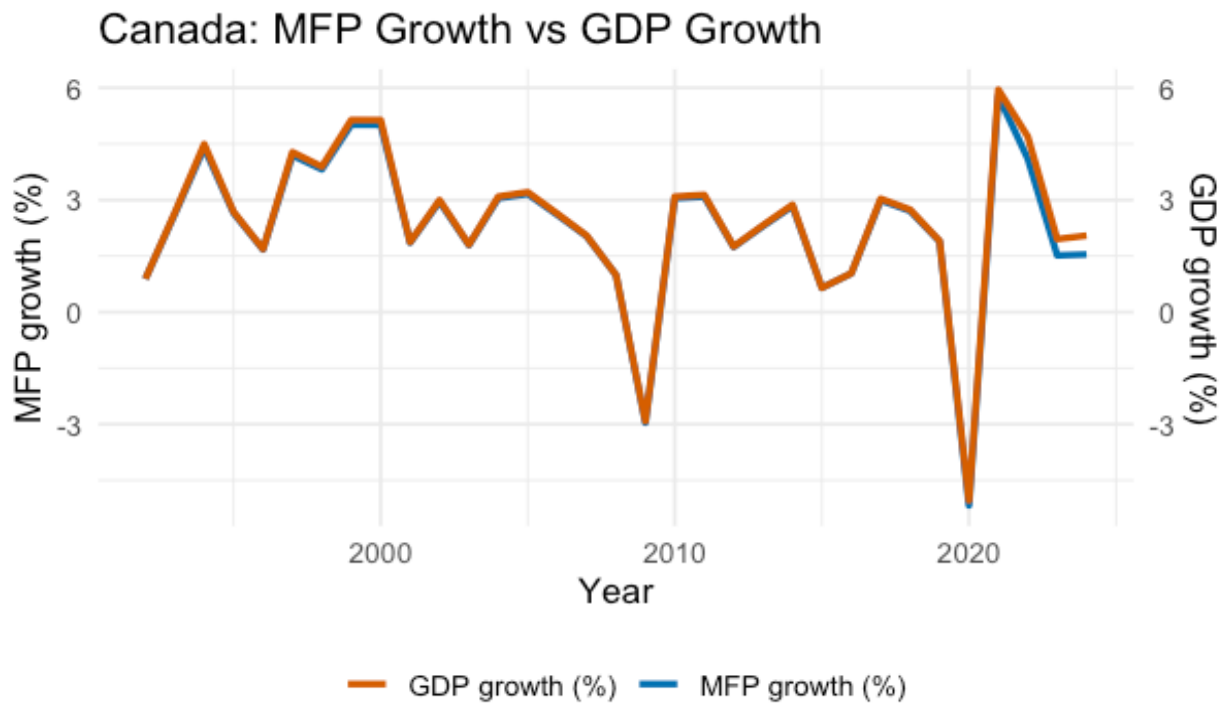
Note: GDP Growth is scaled by a factor of 10 for dual-axis visualization. Source: IMF World Economic Outlook (2024); Statistics Canada.

Figure A2: Population Growth vs GDP Growth



Note: Source: Statistics Canada.

Figure A3: MFP Growth vs GDP Growth



Note: The near-identical tracking of MFP growth and GDP growth motivates the use of $\Delta \log$ (MFP) as the primary productivity control. Source: Statistics Canada.

B. Baseline Regression Results

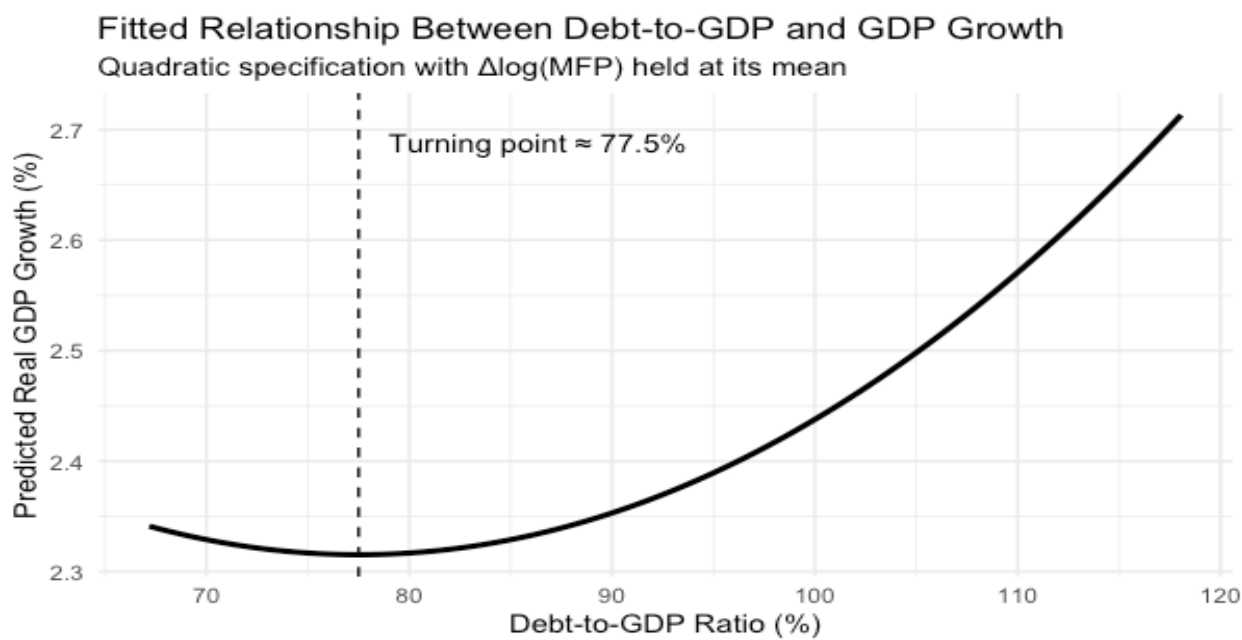
Table B1: Canada — Baseline GDP Growth Regressions (No Lag)

Variable	Linear (1)	Quadratic (2)
Debt-to-GDP (t)	0.006*** (0.002)	-0.037* (0.019)
Debt-to-GDP ² (t)	—	0.0002** (0.0001)

$\Delta \log(\text{MFP})$	1.013*** (0.010)	1.020*** (0.010)
Constant	-0.512*** (0.154)	1.426* (0.832)
Observations	33	33
R ²	0.997	0.997
Adjusted R ²	0.997	0.997
Turning Point (Debt-to-GDP %)	—	77.51%

Note: Dependent variable is Real GDP Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. OLS estimation with Newey-West HAC standard errors.

Figure B1: Fitted Relationship Between Debt-to-GDP and GDP Growth (Baseline)



Note: Quadratic specification with $\Delta \log(\text{MFP})$ held at its sample mean.

C. Lagged Debt Specifications:

Table C1: GDP Growth with 1-Year Lagged Debt-to-GDP

Variable	Coefficient
Lagged Debt-to-GDP (t-1)	-0.038** (0.017)
Lagged Debt-to-GDP ² (t-1)	0.0003** (0.0001)
$\Delta \log(\text{MFP})$	0.991*** (0.009)
Constant	1.433* (0.753)
Observations	33

R ²	0.998
Adjusted R ²	0.998
Turning Point (Debt-to-GDP %)	74.79%

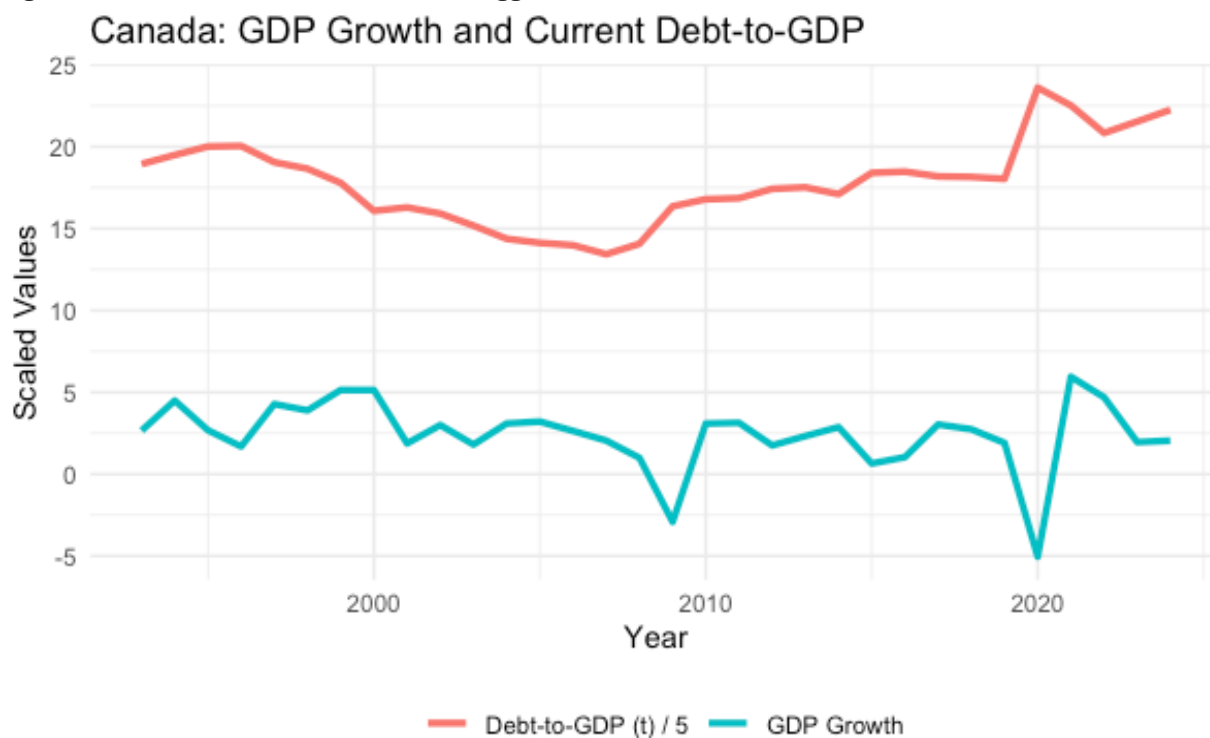
Note: Dependent variable is Real GDP Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

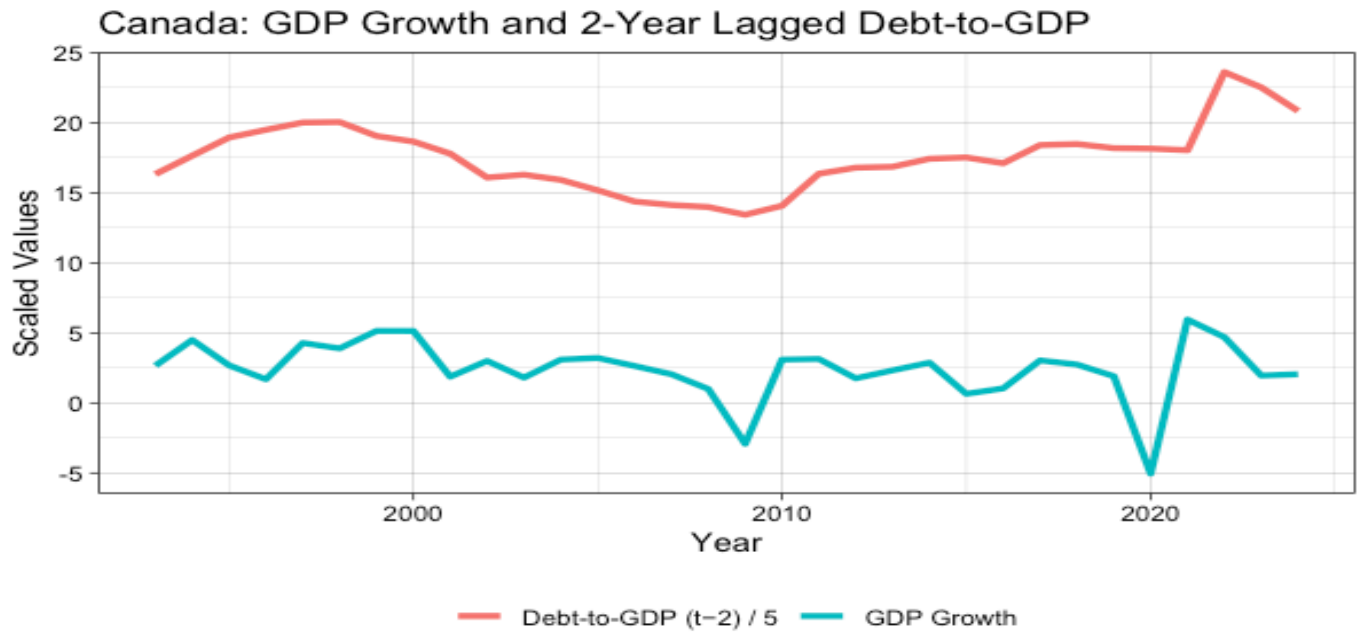
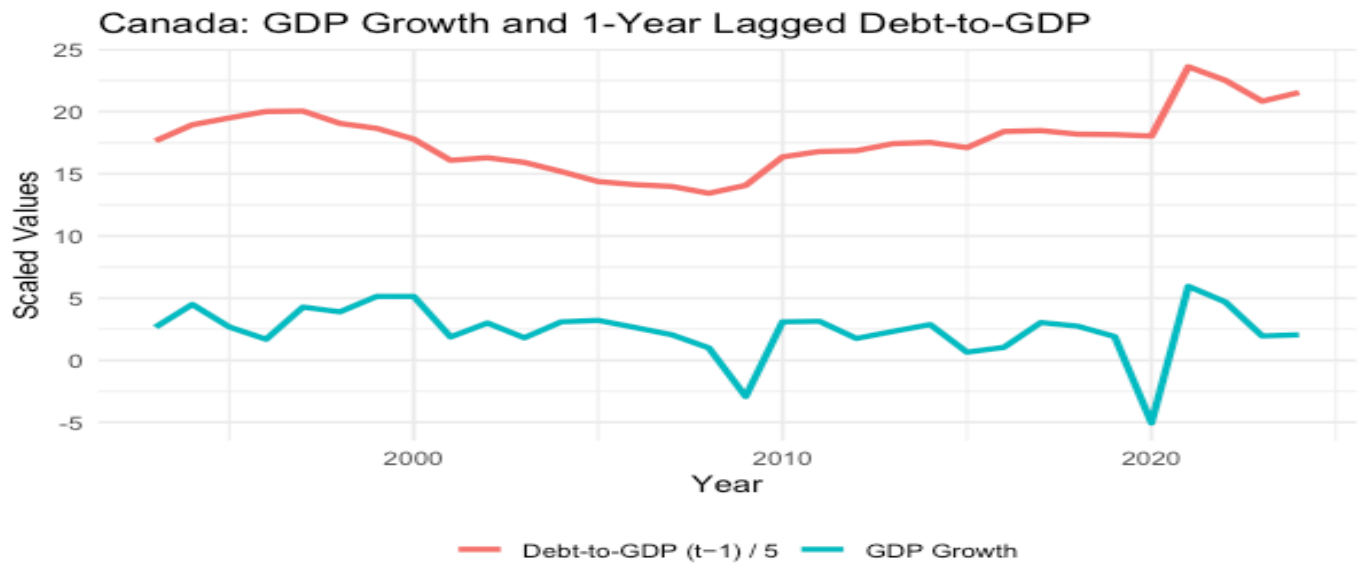
Table C2: GDP Growth with 2-Year Lagged Debt-to-GDP

Variable	Coefficient
Debt-to-GDP (t-2)	-0.061*** (0.012)
Debt-to-GDP ² (t-2)	0.0004*** (0.0001)
$\Delta \log(\text{MFP})$	1.000*** (0.006)
Constant	2.418*** (0.531)
Observations	32
R ²	0.999
Adjusted R ²	0.999
Turning Point (Debt-to-GDP %)	79.12%

Note: Dependent variable is Real GDP Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure C1: GDP Growth and Current vs Lagged Debt-to-GDP





D. State-Dependent Interaction Model:

Table D1: Interactive Model (Lagged Debt, Δ MFP, Crisis Dummies)

Variable	Coefficient
Debt-to-GDP (t-1)	-0.101*** (0.016)
Debt-to-GDP ² (t-1)	0.001*** (0.0001)
$\Delta \log(\text{MFP})$	1.215*** (0.132)
D(2008–09)	-6.665* (3.434)
D(2020–21)	2.487*** (0.698)

Debt($t-1$) $\times$ $\Delta \log(\text{MFP})$	-0.002 (0.001)
Debt($t-1$) $\times$ D(2008–09)	0.099* (0.051)
Debt($t-1$) $\times$ D(2020–21)	-0.025*** (0.007)
Constant	3.817*** (0.761)
Observations	32
R ²	0.999
Adjusted R ²	0.999

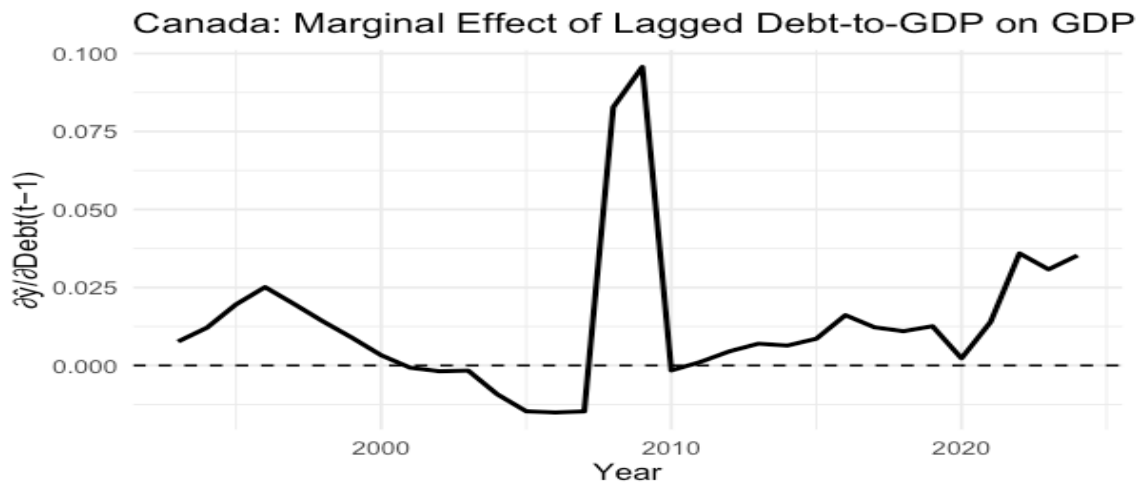
Note: Dependent variable is GDP Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. HAC standard errors used throughout.

Table D2: Marginal Effects of Lagged Debt-to-GDP on GDP Growth

Scenario	Marginal Effect	95% CI Lower	95% CI Upper
Normal year (means, no crisis)	0.0088	0.0036	0.0139
2008–09 crisis year (means)	0.1075	0.0560	0.1589
2020–21 crisis year (means)	-0.0166	-0.0258	-0.0074
Normal year ($\Delta \text{MFP} = 0$; no crisis)	0.0140	0.0047	0.0232

Note: Marginal effects computed using the delta method with HAC standard errors. Evaluated at sample mean of lagged debt-to-GDP.

Figure D1: Marginal Effect of Lagged Debt-to-GDP on GDP Growth (Year-by-Year)



E. GDP per Capita Regressions:

Table E1: GDP per Capita Growth — Baseline Model

Variable	Base Model
Debt-to-GDP	0.202*** (0.052)

Debt-to-GDP ²	-0.001*** (0.0003)
$\Delta \log(\text{MFP})$	1.032*** (0.025)
Dummy: 2008–2009	-0.003 (0.209)
Dummy: 2020–2021	1.893*** (0.279)
Constant	-9.049*** (2.246)
Observations	33
R ²	0.989
Adjusted R ²	0.987
Turning Point (Debt-to-GDP %)	80.04%

Note: Dependent variable is GDP per Capita Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table E2: GDP per Capita Growth — 1-Year Lagged Debt

Variable	Coefficient
Debt-to-GDP (t–1)	0.122* (0.064)
Debt-to-GDP ² (t–1)	-0.001** (0.0004)
$\Delta \log(\text{MFP})$	1.142*** (0.034)
Dummy: 2008–2009	0.435 (0.309)
Dummy: 2020–2021	1.364*** (0.284)
Constant	-6.018** (2.923)
Observations	32
R ²	0.985
Adjusted R ²	0.982

Note: Dependent variable is GDP per Capita Growth (%). Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

F. Diagnostic and Robustness Tests:

Table F1: Baseline Regression Diagnostic Tests

Test	Statistic	df	p-value
Joint F-Test on Debt Terms	F = 3.855	(2, 29)	0.033*

RESET Test (linear)	0.034	(2, 28)	0.967
RESET Test (quadratic)	0.769	(2, 27)	0.473
Breusch-Pagan (heteroskedasticity)	BP = 12.966	3	0.005***
Breusch-Godfrey (serial correlation)	LM = 4.373	2	0.112

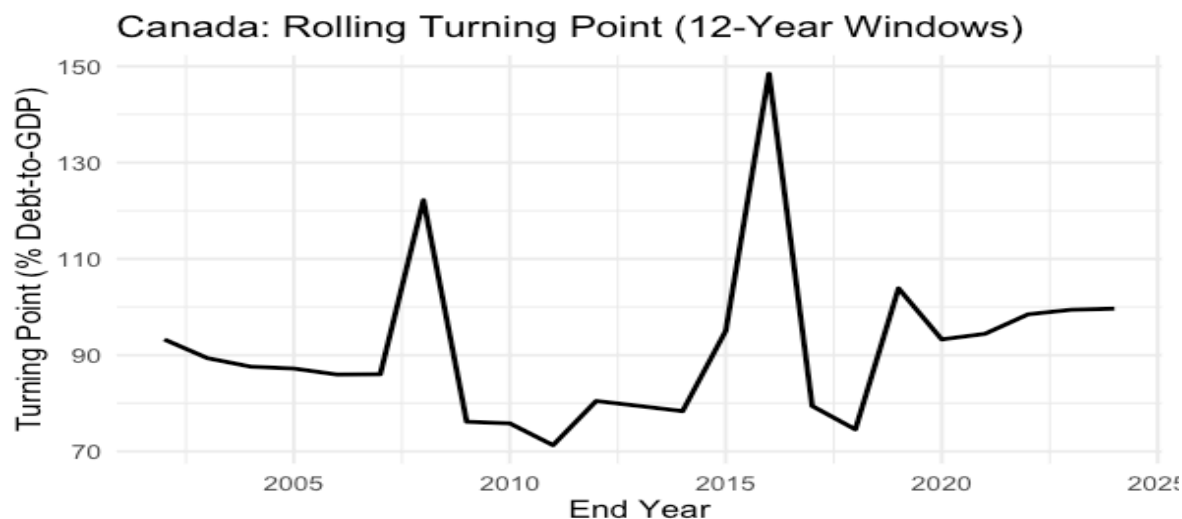
Note: RESET = Ramsey RESET test for functional form misspecification. Breusch-Pagan tests for heteroskedasticity. Breusch-Godfrey tests for serial correlation up to order 2. Significant heteroskedasticity motivates the use of HAC standard errors throughout. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table F2: HAC Inference — Baseline Quadratic Model

Variable	Estimate	HAC Std. Error	t-value	p-value
Constant	1.426	0.837	1.703	0.099.
Debt-to-GDP	-0.037	0.020	-1.888	0.069.
Debt-to-GDP ²	0.00024	0.00012	2.080	0.046*
$\Delta \log(\text{MFP})$	1.020	0.007	140.8	<0.001***

Note: Newey-West HAC standard errors. Under HAC inference, the linear debt term loses significance at conventional levels while the quadratic term retains significance, consistent with a nonlinear association. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

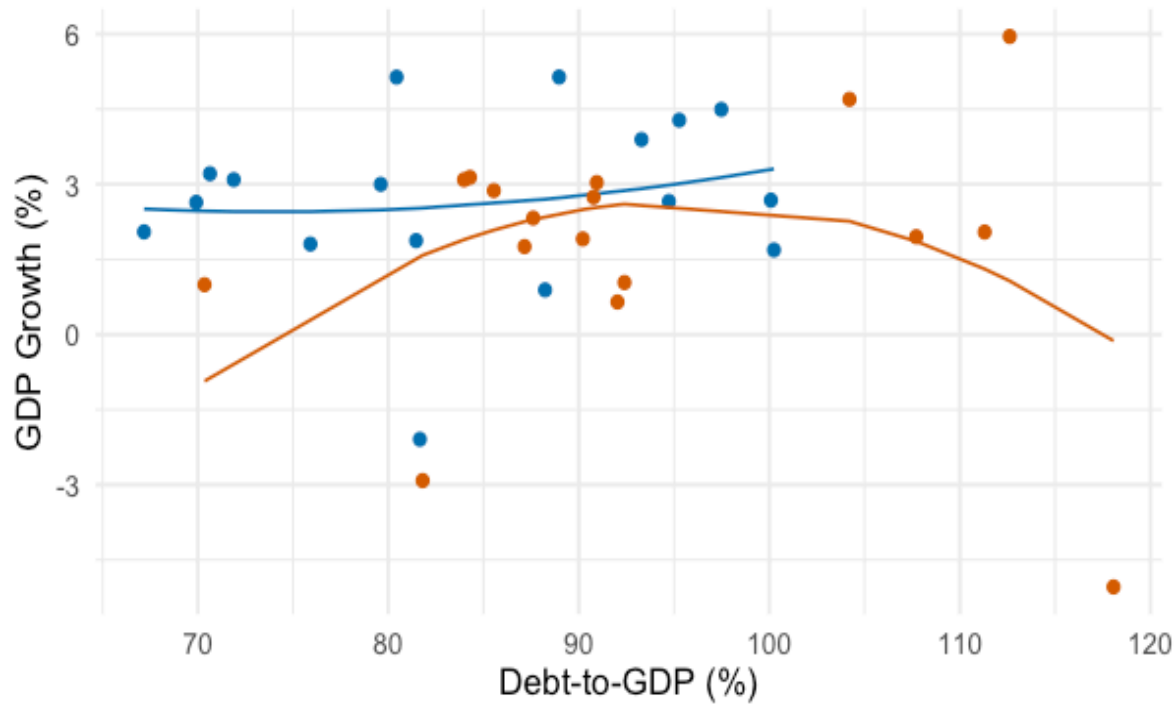
Figure F1: Rolling 12-Year Turning Point Analysis



Note: Rolling window estimates of the quadratic debt turning point. Instability across windows reflects regime changes and small subsample sizes.

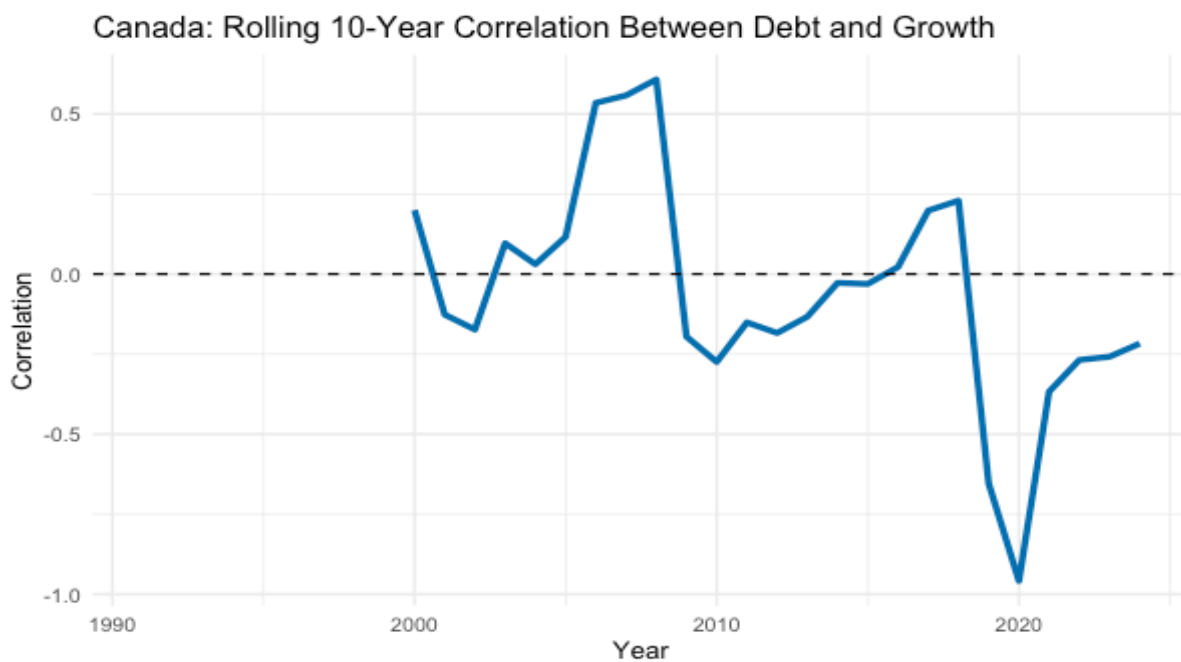
Figure F2: Debt-Growth Curve Before vs After 2008

Debt–Growth Curve Before vs After 2008



Note: Blue = pre-2008 period; Orange = post-2008 period. The post-2008 curve exhibits an inverted U-shape, contrasting with the flatter pre-2008 relationship.

Figure F3: Rolling 10-Year Correlation Between Debt and GDP Growth



Note: The rolling correlation is unstable and changes sign across periods, providing further evidence that the debt-growth relationship is state-dependent rather than structural.

G. Stationarity Tests (Augmented Dickey-Fuller):

Table G1: ADF Test Results — Debt-to-GDP Ratio

Specification	Test Stat (τ)	Critical Value 5%	Reject H_0 ?	Inference
With drift	-0.996	-2.93	No	I(1)
With trend	-1.559	-3.50	No	I(1)
Without trend/drift	-0.650	-1.95	No	I(1)

Note: H_0 = unit root present. Failure to reject in all specifications indicates the debt-to-GDP series is non-stationary (integrated of order 1). This is acknowledged as a limitation in Section III.E.

G. References:

Rehn, O. (2013, July). *Challenges to monetary policy in the euro area*. European Commission.
https://ec.europa.eu/commission/presscorner/detail/en/SPEECH_13_207

Reinhart, C. M., & Rogoff, K. S. (2010). Growth in a time of debt. *American Economic Review*, 100(2), 573–578. <https://doi.org/10.1257/aer.100.2.573>

Department of Finance Canada. (2024). *Budget 2024: Fairness for every generation*. Government of Canada.
<https://budget.canada.ca/2024/home-accueil-en.html>

Panizza, U., & Presbitero, A. F. (2014). Public debt and economic growth: Is there a causal effect? *Journal of Macroeconomics*, 41, 21–41. <https://doi.org/10.1016/j.jmacro.2014.03.009>

Herndon, T., Ash, M., & Pollin, R. (2014). Does high public debt consistently stifle economic growth? A critique of Reinhart and Rogoff. *Cambridge Journal of Economics*, 38(2), 257–279.
<https://doi.org/10.1093/cje/bet075>

Checherita-Westphal, C., & Rother, P. (2012). The impact of high government debt on economic growth and its channels: An empirical investigation for the euro area. *European Economic Review*, 56(7), 1392–1405.
<https://doi.org/10.1016/j.eurocorev.2012.06.007>

International Monetary Fund. (2024). *World Economic Outlook database*. IMF.
<https://www.imf.org/en/Publications/WEO/weo-database>

Statistics Canada. (2024). *Table 36-10-0222-01: Gross domestic product, expenditure-based, Canada*.
<https://www150.statcan.gc.ca>